

BANGLADESH

Historical Climate Data Report

Climatology 1901–2020 | CRU Dataset | World Bank CCKP

Generated: 03 May 2026 | Source: World Bank Climate Change Knowledge Portal (CCKP) — climateknowledgeportal.worldbank.org

1. Country Climate Overview

Bangladesh is situated in South Asia between latitudes 20°34'N – 26°38'N and longitudes 88°01'E – 92°41'E, covering approximately 147,570 km². The country lies at the confluence of three great river systems — the Ganges, Brahmaputra, and Meghna — draining into the Bay of Bengal. This geography makes Bangladesh one of the most climate-vulnerable nations on Earth, exposed to tropical cyclones, flooding, storm surges, and sea-level rise.

The climate is classified as humid subtropical (Köppen: Cwa/Cfa) with a dominant South Asian Monsoon signal. Three main seasons are recognised: a hot dry pre-monsoon (March–May), a hot wet monsoon (June–October), and a cool dry winter (November–February).

Climate Classification	Humid Subtropical (Cwa / Cfa) — Köppen-Geiger
Mean Annual Temperature	~26 °C (range 15 °C – 34 °C seasonally)
Mean Annual Precipitation	~2,300 mm (west ~1,000 mm; east/south >2,800 mm)
Monsoon Contribution	~80 % of annual rainfall in Jun–Oct
Dominant Hazards	Cyclones, riverine/coastal floods, storm surge, drought
Historical Dataset	CRU TS v4 — University of East Anglia; 0.5°×0.5° grid
Reference Climatology	1991–2020 (WMO standard period)
Historical Record Span	1901–2020 (120 years)

2. Temperature — Historical Climatology

2.1 Annual & Seasonal Means

The CRU dataset records a long-term mean annual temperature of approximately 25.8 °C for Bangladesh (1991–2020 baseline). Spatial variability is moderate: the northwest (Rajshahi Division) tends to be hottest and driest, while the Chittagong Hill Tracts in the southeast experience cooler temperatures due to elevation.

Season / Period	Mean Temp (°C)	T-max (°C)	T-min (°C)	Notes
Annual average	25.8	30.5	20.9	Baseline 1991–2020
Pre-monsoon (MAM)	27.3	33.8	20.8	Peak heat; drought risk
Monsoon (JJAS)	28.1	31.5	24.7	Warm and humid

Post-monsoon (ON)	26.0	29.7	22.3	Transition; cyclone season
Winter (DJF)	18.0	25.2	12.8	Cool & dry; frost in north

2.2 Long-term Temperature Trend (1901–2020)

Analysis of the CRU time series shows a clear warming trend, especially from the 1980s onward. Bangladesh's mean temperature has risen by approximately +0.5 °C to +1.0 °C over the 20th century, consistent with global warming patterns. The most pronounced warming is observed during the winter (December–February) season and in night-time minimum temperatures, which have increased at a faster rate than daytime maxima.

Variable	Trend (1901–2020)	Confidence	Most Affected Season
Mean Temperature	+0.8 °C / century	High	Winter (DJF)
Maximum Temperature	+0.6 °C / century	High	Pre-monsoon (MAM)
Minimum Temperature	+1.1 °C / century	High	Winter (DJF)
Diurnal Temp Range	Narrowing trend	Medium	Winter

3. Precipitation — Historical Climatology

3.1 Spatial & Seasonal Distribution

Bangladesh receives 70–80 % of its annual rainfall during the June–September monsoon. The spatial pattern is strongly asymmetric: the Sylhet region (northeast) and Chittagong Division (southeast) receive over 3,000 mm annually, while the northwest (Barind Tract in Rajshahi) receives as little as 1,000–1,200 mm. This west-to-east gradient is driven by the Bay of Bengal moisture flux and orographic effects of the Meghalaya Plateau.

Region / Season	Mean Precip (mm)	% of Annual	Key Risk
Annual (national avg)	2,300	100 %	—
Monsoon (Jun–Sep)	1,600	~70 %	Riverine flooding
Pre-monsoon (Mar–May)	380	~17 %	Nor'westers / thunderstorms
Post-monsoon (Oct–Nov)	200	~9 %	Cyclone-enhanced rainfall
Winter (Dec–Feb)	60	~3 %	Drought risk in northwest
Sylhet / NE (annual)	3,500+	—	Highest in country
Rajshahi NW (annual)	1,100	—	Driest region

3.2 Long-term Precipitation Trend (1901–2020)

Unlike temperature, precipitation trends in Bangladesh are more complex and regionally variable. The CRU time series suggests a slight increase in overall annual precipitation since the mid-20th century, but with significant interannual variability driven largely by the El Niño–Southern Oscillation (ENSO). La Niña years are typically associated with above-normal monsoon rainfall and increased flood risk, while El Niño years tend towards below-normal rainfall and drought conditions.

There is evidence of intensification of extreme precipitation events even as some regions experience longer dry spells — a 'wet gets wetter, dry gets drier' pattern consistent with global warming projections for the tropics.

4. Extreme Climate Events

4.1 Tropical Cyclones

The Bay of Bengal accounts for roughly 7 % of global tropical cyclone activity but disproportionately affects Bangladesh due to its funnel-shaped coastline and shallow continental shelf that amplifies storm surge. Historically, the country experiences 1–2 significant cyclone landfalls per decade. Notable events include the Bhola Cyclone (1970, ~300,000 fatalities), Cyclone Sidr (2007), and Cyclone Amphan (2020).

4.2 Floods

Approximately 20–30 % of Bangladesh's land area floods annually under normal monsoon conditions, rising to over 60 % during severe flood years (e.g., 1988, 1998, 2004, 2017). Floods stem from three main sources: (i) overbank discharge from the Ganges-Brahmaputra-Meghna system, (ii) direct heavy rainfall runoff, and (iii) coastal/tidal flooding exacerbated by cyclone storm surge.

4.3 Droughts & Heat Waves

Northwest Bangladesh (Rajshahi and Rangpur Divisions) faces recurring agricultural drought during the pre-monsoon and dry winter season. Increasing temperatures and more variable pre-monsoon rainfall are extending the effective drought period. Heat wave frequency and intensity are also increasing, with the pre-monsoon period recording maximum temperatures exceeding 40 °C in recent decades.

Hazard	Frequency	Trend	Most Affected Area
Tropical Cyclones	1–2 per decade (major)	Intensity ↑	Coastal divisions
Riverine Floods	Annual (20–30 % area)	Frequency uncertain	All river basins
Flash Floods	Irregular	Increasing	Sylhet, CHT
Storm Surge	With cyclones	Severity ↑	Coastal belt
Agricultural Drought	~1 in 3–4 years	Duration ↑	NW Bangladesh
Heat Waves	Pre-monsoon, annual	Intensity ↑	NW/Central

5. Interannual Variability & ENSO

The primary driver of year-to-year variability in Bangladesh's climate is the El Niño–Southern Oscillation (ENSO). During El Niño phases, the South Asian Monsoon tends to be suppressed, resulting in below-normal rainfall and increased drought risk. La Niña phases are associated with above-normal monsoon rainfall and enhanced flood probability. The Indian Ocean Dipole (IOD) also modulates monsoon intensity, with positive IOD events generally enhancing rainfall over South Asia.

Decadal-scale oscillations — including the Pacific Decadal Oscillation (PDO) and the Atlantic Multidecadal Oscillation (AMO) — further modulate the long-term variability, making attribution of trends to anthropogenic

climate change more challenging at sub-centennial timescales.

6. Data Source & Methodology

The historical data presented in this report is drawn from the **Climatic Research Unit (CRU) TS v4** dataset, produced by the University of East Anglia (UEA). Key dataset characteristics:

- Spatial resolution: $0.5^\circ \times 0.5^\circ$ (~50 km × 50 km grid)
- Temporal coverage: 1901 to present (monthly means)
- Variables: Mean temperature (TMP), Maximum temperature (TMX), Minimum temperature (TMN), Precipitation (PRE), Vapour pressure, Cloud cover, Wet-day frequency
- Station basis: Compiled from global land-surface weather stations; interpolated using angular distance weighting
- Reference climatology: 1991–2020 (WMO standard normal period)
- Portal: World Bank Climate Change Knowledge Portal — climateknowledgeportal.worldbank.org

Caveat: In regions with sparse weather-station coverage — particularly during the early 20th century — CRU data quality may be affected by spatial and temporal gaps, leading to potential biases. For more detailed analysis, the ERA5 reanalysis dataset (available on the same portal) is recommended.

7. Key Findings & Implications

- **Warming trend confirmed:** Mean temperatures have risen ~0.8 °C over the 20th century, with the fastest warming in winter minimum temperatures (+1.1 °C/century).
- **Monsoon dominance:** ~70–80 % of annual rainfall arrives in Jun–Sep; small shifts in monsoon onset or intensity have major consequences for agriculture and floods.
- **Extreme rainfall intensifying:** Evidence of heavier individual rain events even while some dry-season periods are becoming drier — increasing both flood and drought risk.
- **Cyclone threat:** While cyclone frequency may not increase dramatically, intensity and associated storm surge are projected to worsen with rising sea-surface temperatures.
- **Sea-level rise amplifier:** Bangladesh's low-lying delta is acutely sensitive to even small changes in mean sea level, which amplify coastal flooding and salinity intrusion.
- **Adaptation priority:** Improved early-warning systems, climate-resilient agriculture, coastal embankments, and urban heat-island mitigation are critical adaptation priorities.